HOTEL BOOKING DEMAND ANALYSIS

DATA PREPROCESSING REPORT

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Dataset Overview

Metric	Value
Total Records	87230
Total Features	46
Missing Values	0
Duplicate Records	0

Project Overview

This project focuses on analyzing hotel booking demand using data preprocessing, exploratory data analysis (EDA), statistical analysis, hypothesis testing, and an interactive Flask dashboard. The objective of the preprocessing stage is to improve the quality of the dataset by handling missing values, removing duplicates, correcting data types, treating outliers, and creating meaningful features for further analysis.

Data Cleaning Process

1. Missing Values

The dataset was examined for missing values. Missing values were identified using pandas 'isnull()' function. Categorical variables were imputed using the mode, while numerical variables were handled using appropriate statistical techniques such as median imputation where necessary.

2. Duplicate Records

Duplicate records were identified using the duplicated() function and removed to maintain dataset consistency and prevent biased analytical results.

3. Data Type Conversion

Date columns were converted into datetime format to enable time-based analysis. Categorical variables were encoded where necessary for statistical analysis and machine learning compatibility.

4. Outlier Treatment

Outliers in the Average Daily Rate (ADR) were detected using boxplots and Z-score analysis. Extreme values were examined and treated appropriately to improve the quality of statistical analysis.

Feature Engineering

Several new features were created to enhance business understanding and analytical capability. • Total Stay • Total Guests • Lead Time Category • Stay Type • Revenue Category • Season • Guest Status These engineered features helped improve visualization, statistical analysis and business insight generation.

Conclusion

The preprocessing stage successfully transformed the raw hotel booking dataset into a clean, consistent, and analysis-ready dataset. The cleaned dataset was subsequently used for exploratory data analysis, statistical testing, dashboard development, and business insight generation.